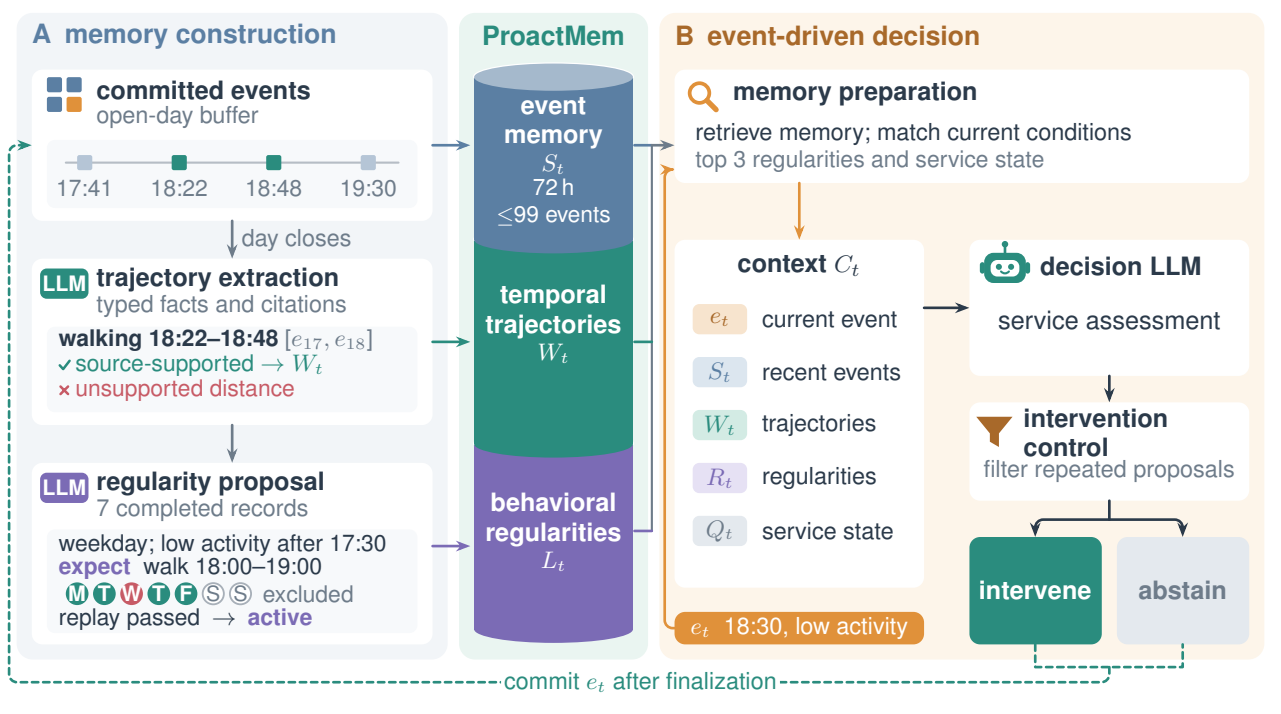

A memory construction
committed events
open-day buffer
17:41
18:22
18:48
19:30
day closes
LLM
trajectory extraction
typed facts and citations
walking 18:22–18:48 [e_{17}, e_{18}]
✓ source-supported → W_t
× unsupported distance
LLM
regularity proposal
7 completed records
weekday; low activity after 17:30
expect walk 18:00–19:00
M T W T F S S excluded
replay passed → active
ProactMem
event memory
S_t
72 h
≤99 events
temporal trajectories
W_t
behavioral regularities
L_t
B event-driven decision
memory preparation
retrieve memory; match current conditions
top 3 regularities and service state
context C_t
e_t current event
S_t recent events
W_t trajectories
R_t regularities
Q_t service state
decision LLM
service assessment
intervention control
filter repeated proposals
intervene
abstain
e_t 18:30, low activity
commit e_t after finalization